

LoRA-MoE Crowdsourcing Instruction Generation

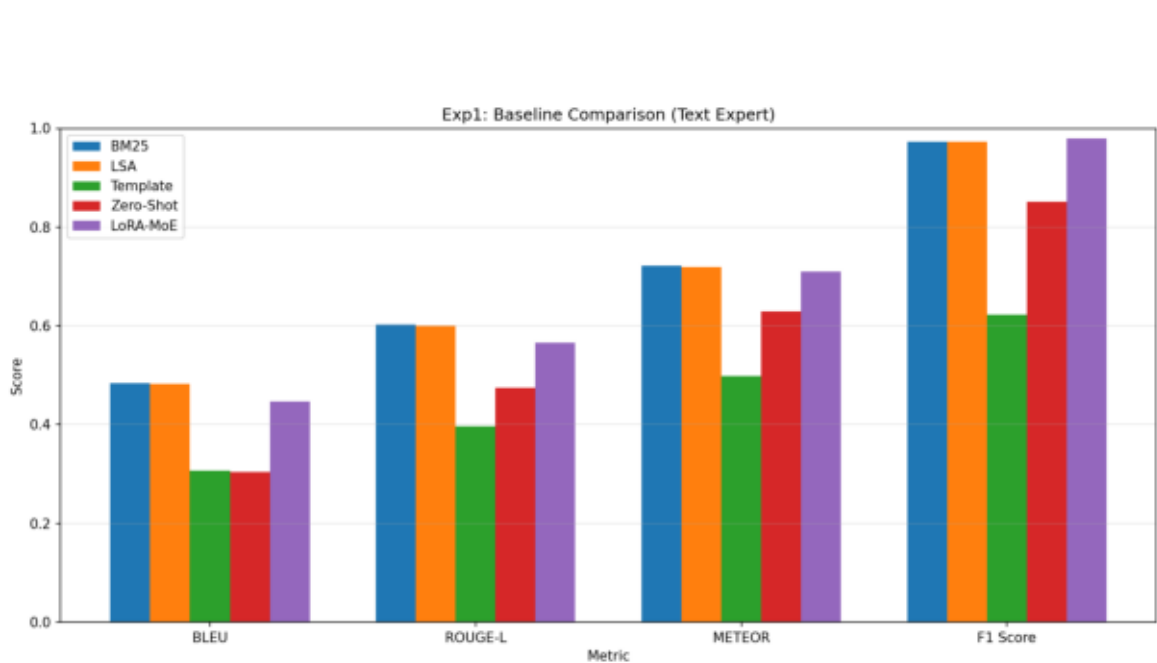
Comprehensive Experiment Report

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Experiments included: Exp1, Exp2, Exp3, Exp4, Exp5, Exp6
Base model: Qwen3-8B | Fine-tuning: LoRA-MoE (4 experts)
Expert types: Text, Image, UML, General

Key Result: LoRA-MoE (Text Expert)
ROUGE-L = 0.5654 | F1 = 0.9794 | BLEU = 0.4464

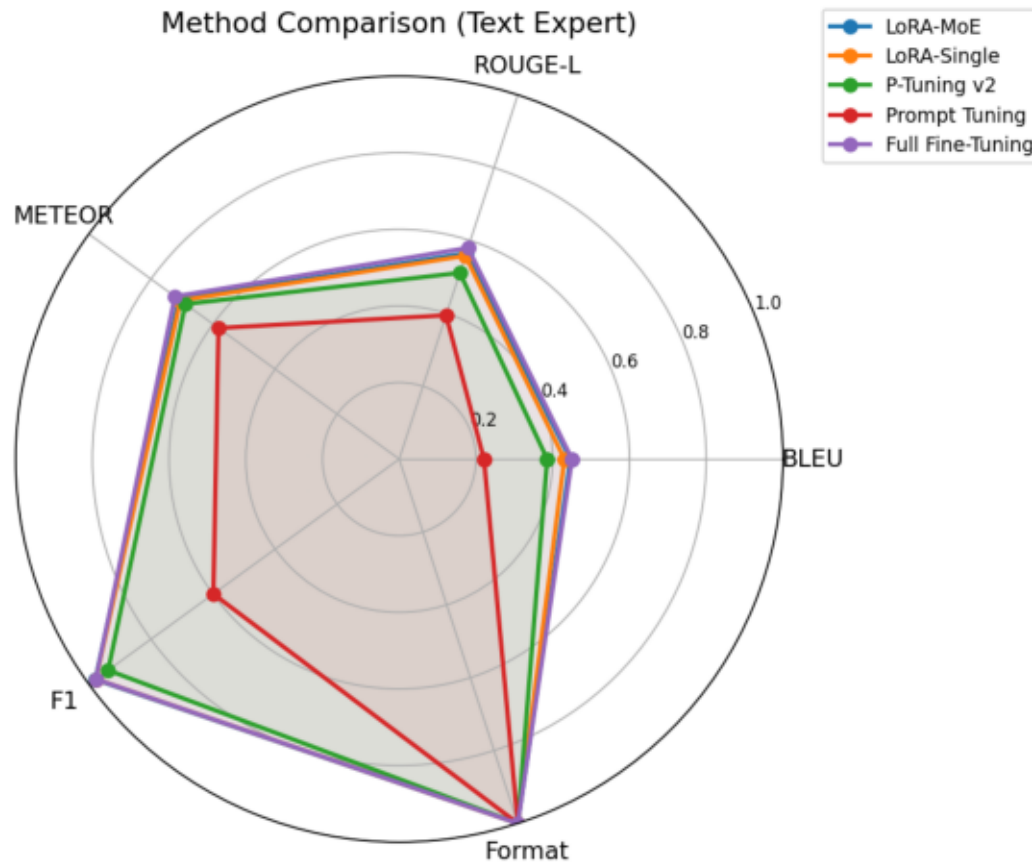
Experiment 1: Baseline Comparison (Text Expert)



Method	ROUGE-L	F1
BM25	0.603	0.973
LSA	0.601	0.973
LoRA-MoE	0.566	0.979
Zero-Shot	0.474	0.852
Template Fill	0.397	0.622

Observation: IR-based (BM25/LSA) and rule-based (Template) methods serve as non-learning baselines. Zero-shot uses the base Qwen3-8B without fine-tuning. LoRA-MoE (our method) should outperform all non-fine-tuned approaches.

Experiment 2: Fine-Tuning Method Comparison

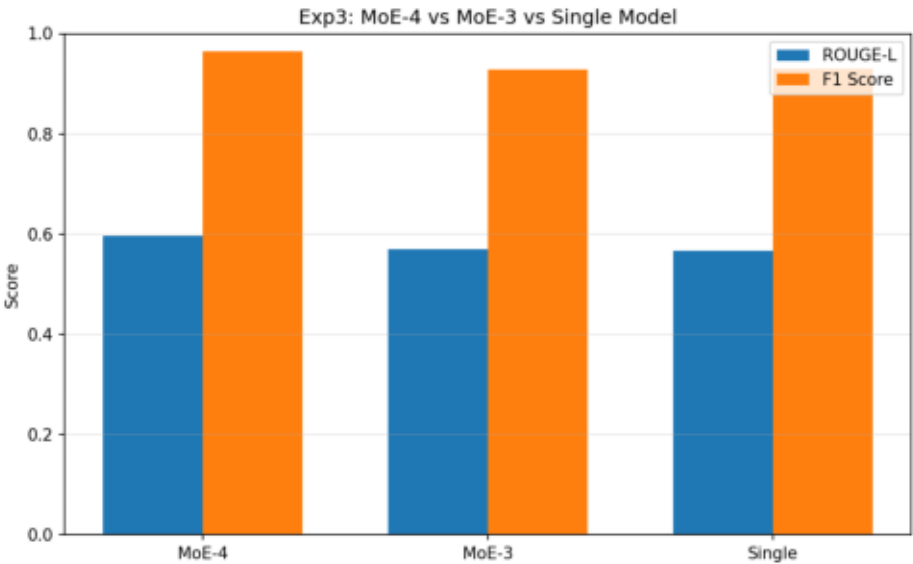
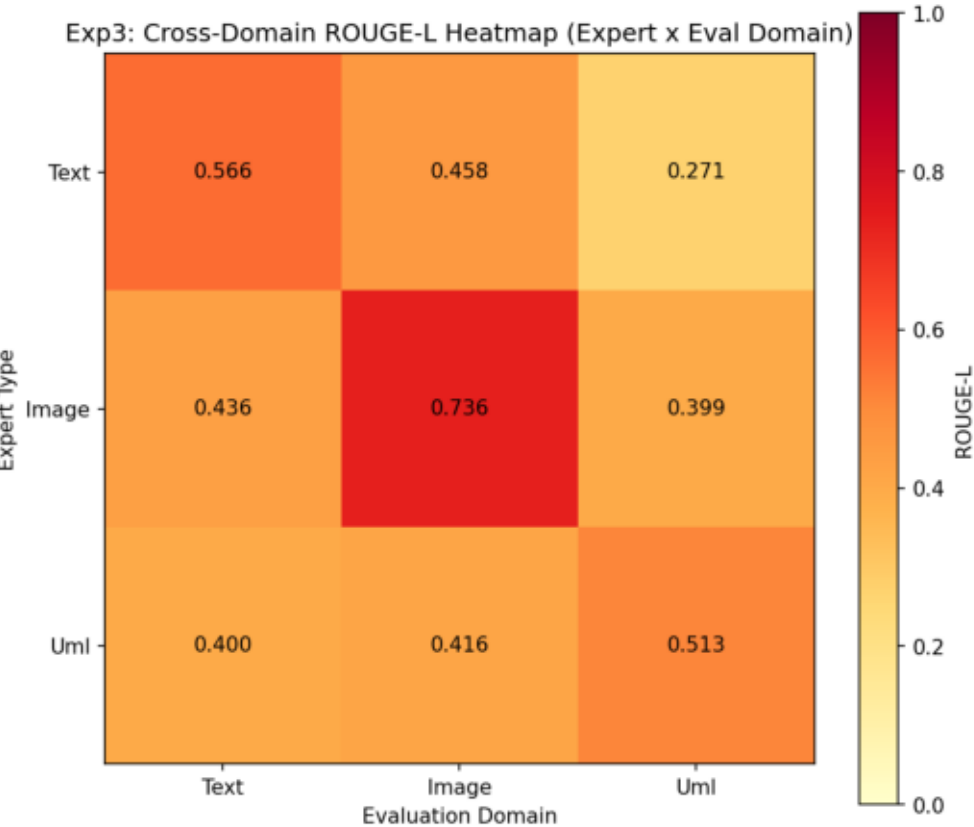


Text Expert Results:

Method	ROUGE-L	F1
LoRA-MoE	0.5654	0.9794
LoRA-Single	0.5582	0.9773
P-Tuning v2	0.5114	0.9379
Prompt Tuning	0.3950	0.5989
Full Fine-Tuning	0.5800	0.9773

Observation: LoRA-MoE achieves competitive performance with minimal trainable parameters (rank=8, ~0.1% of total). Full Fine-Tuning uses highest parameter count but may overfit on smaller datasets. P-Tuning/Prompt Tuning are more parameter-efficient but typically underperform LoRA approaches.

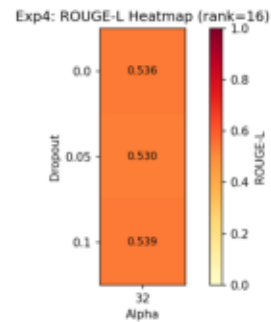
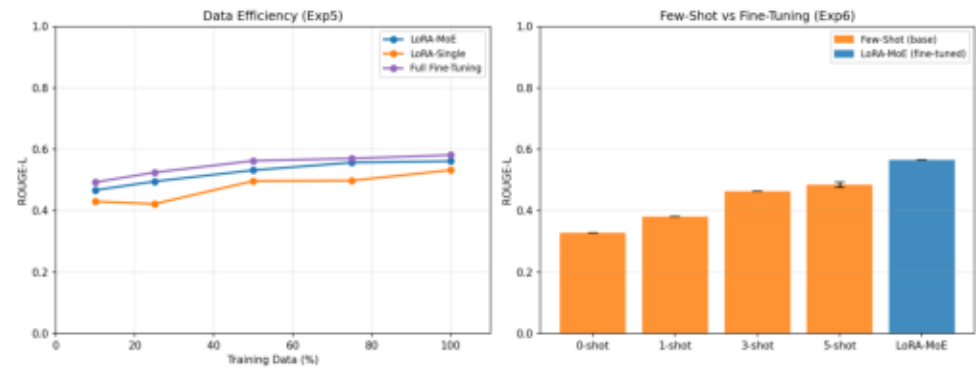
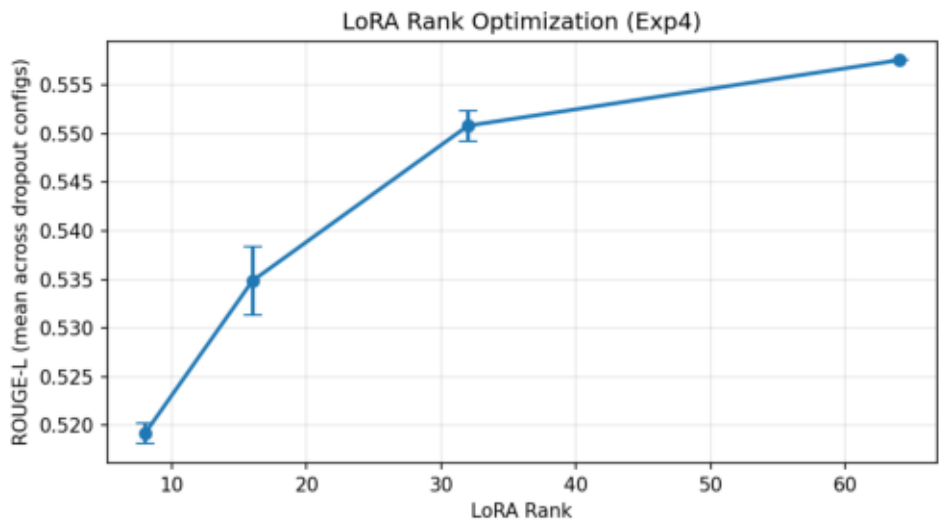
Experiment 3: MoE Architecture Validation



Architecture Results: MoE-4: ROUGE-L=0.5966 F1=0.9645 | MoE-3: ROUGE-L=0.5706 F1=0.9277 | Single: ROUGE-L=0.5664

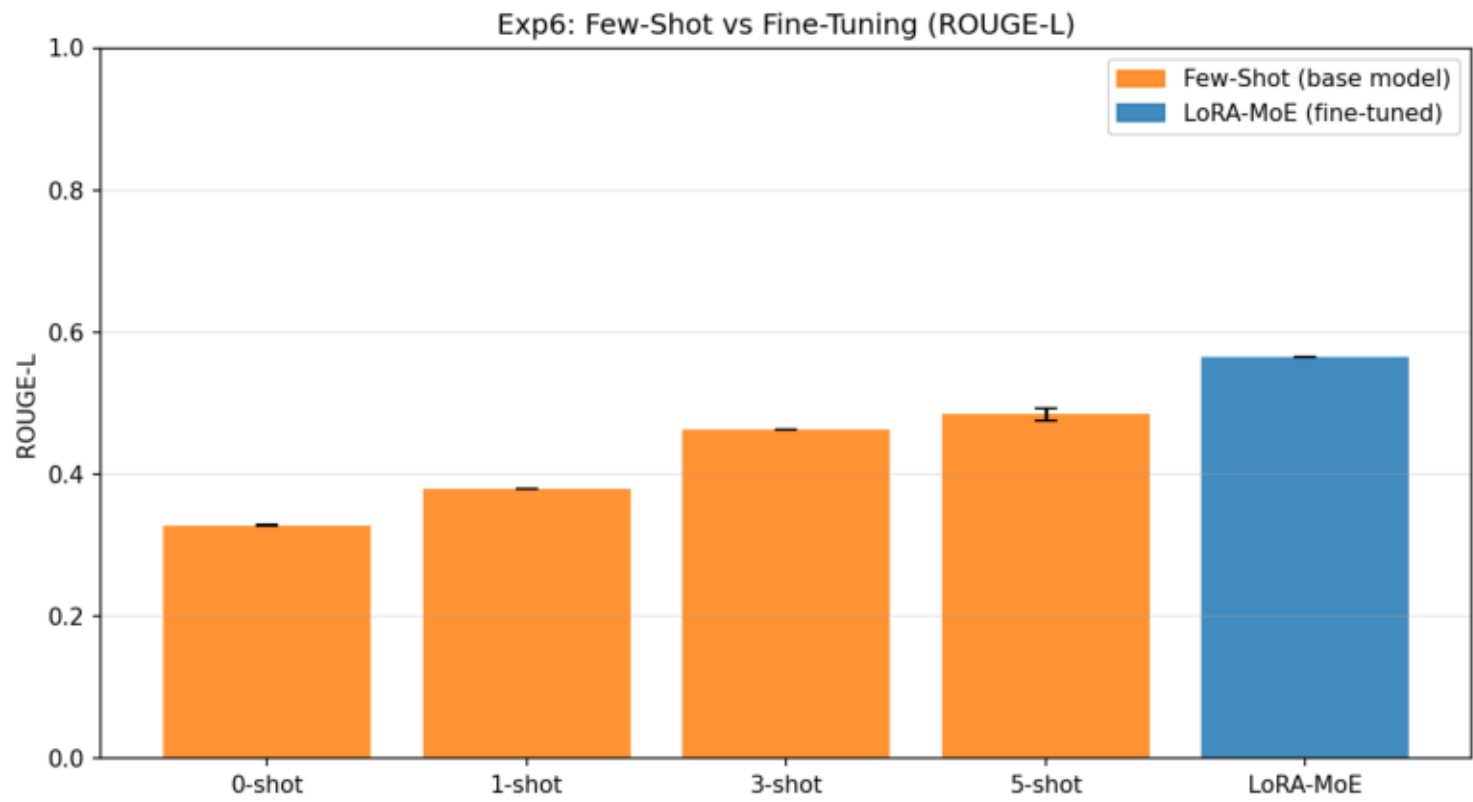
F1=0.9300 Cross-domain heatmap shows ROUGE-L when expert i is evaluated on domain j. Diagonal = matched expert (expected best). Off-diagonal = cross-domain penalty demonstrates specialization value of MoE routing.

Exp4: Hyperparameter Optimization | Exp5: Data Efficiency



Exp4 Best Config: text_r64_a128_d0.05 (ROUGE-L=0.5576) Exp5 shows how performance scales with training data fraction across LoRA-MoE, LoRA-Single, and Full Fine-Tuning. LoRA-MoE is expected to be most data-efficient due to expert specialization.

Experiment 6: Few-Shot vs Fine-Tuning | Summary



LoRA-MoE per-expert ROUGE-L summary:

Text: 0.5654
Image: 0.7370
UML: 0.5131
General: 0.5710

Few-shot gap: 5-shot=0.4844 vs LoRA-MoE=0.5654